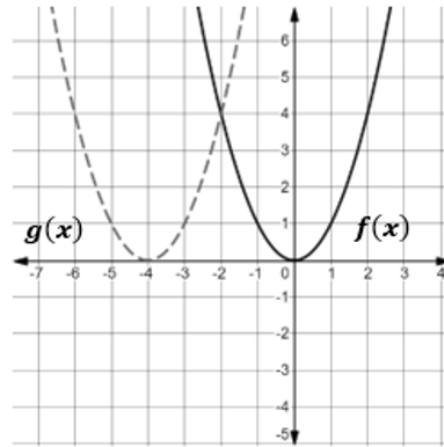


Algebra 1
Unit 13
Lesson 5 Practice

Name _____
Date _____
Period _____

The graphs of $y = f(x)$ and $y = g(x)$ are shown.



- Describe how to shift the graph of $f(x)$ so that it coincides with the graph of $g(x)$.
- Use the graph to complete the table of values for $f(x)$ and $g(x)$.

$f(x)$		$g(x)$	
x	y	x	y
-1		-5	
0		-4	
1		-3	

- Explain the change in the x -values from $f(x)$ to $g(x)$.

- Complete the statement:

The graph of $g(x)$ is a
☐ horizontal
☐ vertical
 shift of $f(x)$.

- Complete the statement:

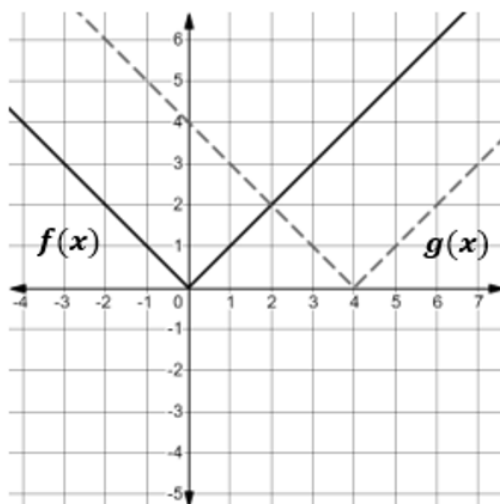
The shift is seen in the table by
☐ adding
☐ subtracting
 4 to each
☐ x -value
☐ y -value

 for each corresponding
☐ x -value.
☐ y -value.

Complete the table for questions 6-8. The first row has been completed.

Original Equation	Original Function	Shift	Transformed Function
$y = x^2$	$f(x) = x^2$	Right 2	$f(x - 2) = (x - 2)^2$
6. $y = x $		Right 4	
7. $y = x^3$		Left 4	
8. $y = \sqrt{x}$		Right 5	

The graphs of $y = f(x)$ and $y = g(x)$ are shown.



9. How does the graph of $f(x)$ compare to the graph of $g(x)$?
10. Two function descriptions are shown. Which function description matches the transformation on the graph? Justify your choice.
- ☐ $g(x) = f(x + 4)$
 - ☐ $g(x) = f(x - 4)$